Instant World Creation

From Text to 3D in Under One Second

From Generative Priors to Feed-Forward Geometry

Deep Learning Final Project · Powered by Apple SHARP

1. The Bottleneck

Photogrammetry

The first traditional approach to 3D reconstruction: scanning from the real world.

- **Principle:** Reconstruct 3D models from multi-angle photographs
- **Requirements:**
  - Physical object must exist
  - Complex camera rigs + controlled lighting
- **Pain Points:**
  - **Cannot scan things that don't exist**
  - Not suitable for creative content generation

NeRF & Optimization-based 3DGS

Optimization Methods: Every scene requires "retraining"

Pain Point: Every new object requires retraining → Cannot scale

The Revolution: Feed-Forward

From "Optimization" to "Prediction"

2. Apple SHARP

Context & Community Impact

Apple SHARP (Sharp Monocular View Synthesis)

Released: Dec 2025 (CVPR 2026 Submission)

- **Community Response:**
  - GitHub Stars: **5.2k+** (first month)
  - Twitter/X: Called "Midjourney for 3D"
- **Why the hype?**
  - First mobile-optimized SOTA model
  - Desktop GPU-level quality

Feed-Forward Architecture

What is a "Feed-Forward" model?

Unlike NeRF (which "memorizes" scenes), SHARP **"understands"** geometry.

```
Image (Pixels) → Neural Network → 3D Gaussians (Geometry)
```

Technical Details: Why So Strong?

1. **Epipolar Constraints**

   SHARP doesn't just guess; it enforces physical laws of vision (epipolar geometry) inside the network layers.

2. **High-Quality Training Data**

   Trained on a curated subset of Objaverse-XL, filtering out low-quality meshes.

3. **Efficient Architecture**

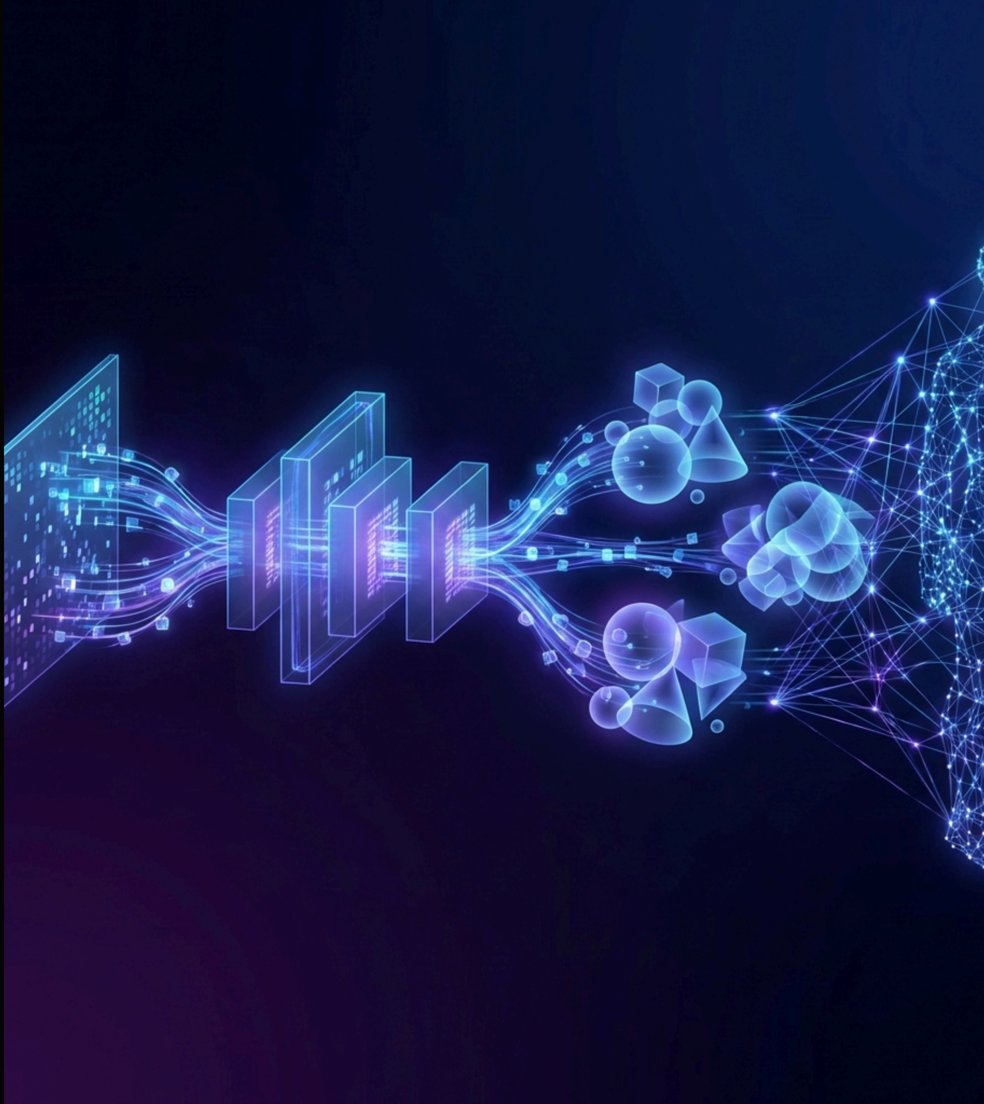
   Mobile-optimized, can run in real-time on iPhone/iPad.

Capabilities & Analogy

"Stable Diffusion for 3D"

Just as **Stable Diffusion** transformed 2D art, **SHARP** transforms 3D assets.

- **Analogy:**
  - **Text-to-Image:** Input "cat", get JPG in 0.5s
  - **Image-to-3D:** Input JPG, get 3D model in 0.2s
- **Key Capabilities:**
  1. **Generalization:** Cats, cars, buildings, abstract art - all work



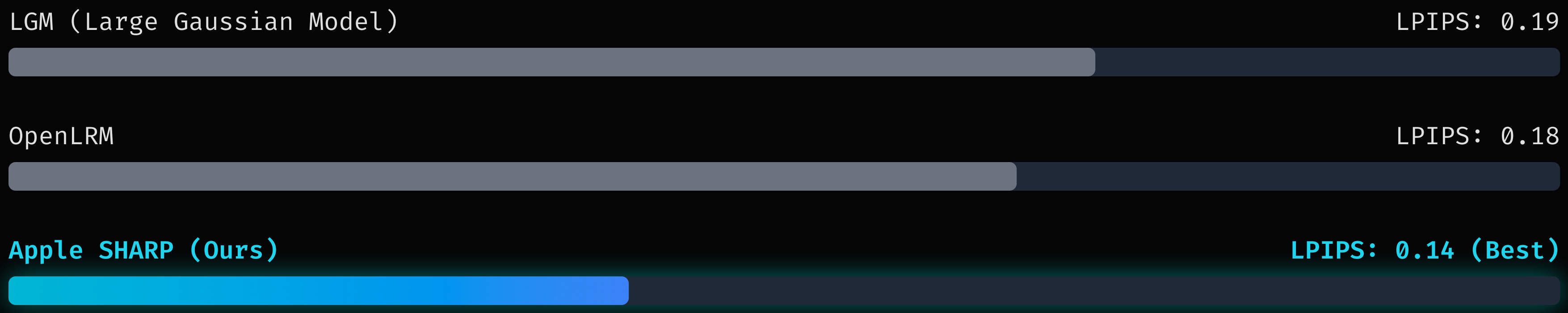
3. Benchmarks

Defining "Quality" for 3D Generative AI

Before looking at data, we must understand the **metrics**.

Comparison Data: Crushing the Competition

Comparison against SOTA methods (LGM, Splatter Image, OpenLRM)



Conclusion: Dominating All Metrics

🏆

SHARP leads in both quality and speed

Enough numbers. Let's see the Product. ↓

4. Engineering Practice

Tech Stack:

- **T2I:** Gemini 2.5 Flash (Nano Banana)
- **3D:** Apple SHARP (Feed-Forward 3DGS)
- **Frontend:** React Three Fiber
- **Backend:** FastAPI + Gradio

```
3d-scene-generator/
├── backend/
│   ├── main.py          # FastAPI
│   ├── app_gradio.py    # HF Space
│   └── services/
│       ├── gemini_service.py
│       └── sharp_service.py
├── frontend/src/
│   ├── GaussianSplatViewer.tsx
│   └── PromptPanel.tsx
└── hf-space/
```

Live Preview: 3D Scene Generator

Our deployed demo on **HuggingFace Space**.

🎨

Live Demo

Visit our deployed application:

🚀 Open 3D Scene Generator

(Click to open in new tab - HF Space requires external access)

Engineering Challenges

Solving the "Illusion" Problem

Problem: Single-view reconstruction creates **"Floaters"** (artifacts) and blurs on the back side.

Solution: We implemented **SceneFog + Camera Bounds**

- Fog fades to black, gracefully hiding edge artifacts
- WASD + Q/E movement with soft boundary constraints

```
// GaussianSplatViewer.tsx
function SceneFog() {
  const { scene } = useThree();

  useEffect(() => {
    // Fog hides black edges
    scene.fog = new THREE.Fog(0x000000, 1.5, 4);
    scene.background = new THREE.Color(0x000000);
    return () => {
      scene.fog = null;
    };
  }, [scene]);
  return null;
}

// Movement bounds
const BOUNDS = {
  minX: -5,
  maxX: 5,
  minY: -3,
  maxY: 3,
  minZ: -5,
  maxZ: 5,
};
```

5. Future Outlook: World Model

From "Object Generation" to "World Generation"

Fei-Fei Li's Vision

World Model = Simulates not just pixels, but **physics and dynamics**

3DGS Applications

Beyond 3D Generation

Future Tech: 4D & Physics

From Static to Dynamic

Current (SHARP)

Thank You

🚀

Deep Learning Final Project

Text-to-3D Instant Generation

>> Google Gemini 2.5 Flash

>> React Three Fiber

>> Apple SHARP

>> FastAPI

github.com/ToadPresident/3d-scene-generator